MPM 2D

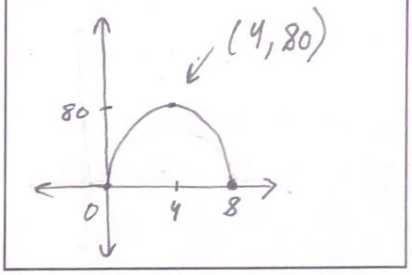
Quadratic Functions - Worksheet #5
Word Problems with $y = a(x - h)^2 + k$

Name: Anna Denisova
Date: Nov. 10, 2022

1. A golf ball is hit and travels through the air according to the equation $h = -5(t - 4)^2 + 80$, where h is the height of the ball in metres, and t is the time in seconds.

x-int
y-int
vertex (max/min)

Rough Sketch (label vertex)



a) The maximum height of the ball, - (y-val vertex)
The max height is 80m.

b) The time it takes to reach the maximum height, - (x-val vertex)
vertex
The time is 4 seconds.

* take furthest one to get where it LANDS, not STARTS (s < e)

c) How long it takes for the ball to land. - (x-int) (the second one)

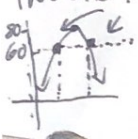
let $h=0$
$0 = -5(t-4)^2+80$
$-80 = -5(t-4)^2$
$16 = (t-4)^2$
$\pm 4 = t-4$

$\pm 4 = t-4$
$4 = t-4$ or $-4 = t-4$
$\boxed{t=8}$ or $\boxed{t=0}$

∴ it hit after 8 seconds.

d) When was the ball 60 m above the ground?
y

two ans:



let $h = 60$
$60 = -5(t-4)^2+80$
$-20 = -5(t-4)^2$
$4 = (t-4)^2$
$\pm 2 = t-4$

$2 = t-4$ $\boxed{t=6}$
$-2 = t-4$ $\boxed{t=2}$

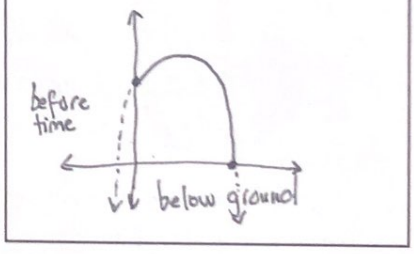
∴ The ball is at 60m after 2 seconds & 6 seconds.

2. A basketball player takes a jump shot. The path of the ball can be described by the equation $h = -5(t - 1)^2 + 7$, where h is the height of the ball in m, and t is the time in seconds. Determine:

V(1, 7)

a) The maximum height of the ball.
vertex
∴ The max height is 7m

Rough Sketch (label vertex)



b) The time it takes to reach the maximum height,
∴ The max height occurs at 1 second.

c) The height from which the player shot the ball.
y-int

let $t = 0$
$h = -5(0-1)^2+7$
$\boxed{h=2}$

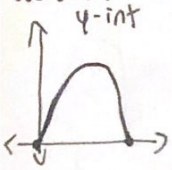
∴ The ball was shot 2m high.

$y = a(x-h)^2 + k$

$V(50, 15000)$

3. A company that builds phones earns a profit on sales according to the quadratic relation $P = -10(N - 50)^2 + 15000$, where P is the profit earned in dollars, and N is the number of crates of phones sold.

we don't know y-int



a) How many crates of phones need to be sold to make a maximum profit?

x

∴ 50 crates need to be sold.

b) What is the maximum profit that can be made?

∴ The max profit is $15,000

Now we know!

c) If the company only sells 11 crates, will they make a profit or take a loss? How much?

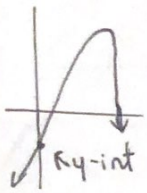
let $n = 11$

$P = -10(11 - 50)^2 + 15000$

$P = -10(-39)^2 + 15000$

$P = -10(1521) + 15000$

$P = -210$



∴ They have a loss of $210.

4. The equation, $h = -0.025(d - 20)^2 + 10$, shows the height of a soccer ball, h metres, as a function of the horizontal distance, d metres, the ball travels until it first hits the ground.

y

x

$V(20, 10)$

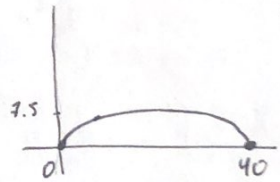
a) What is the maximum height of the ball?

The max height is 10 m.

b) What is the horizontal distance of the ball from the kicker when it reaches its maximum height.

The distance is 20m.

Rough Sketch (label vertex)



c) How far does the ball travel horizontally from when it is kicked until it hits the ground?

let $h = 0$

$0 = -0.025(d - 20)^2 + 10$

$0 = -\frac{1}{40}(d - 20)^2 + 10$

$-10 = -\frac{1}{40}(d - 20)^2$

$400 = (d - 20)^2$

$\pm 20 = d - 20$

$20 = d - 20$ OR $-20 = d - 20$

$\boxed{d = 40}$ OR $\boxed{d = 0}$

∴ The ball traveled 40 m.

d) What is the height of the ball when it is 10 m horizontally from the kicker?

let $d = 10$

$h = -\frac{1}{40}(10 - 20)^2 + 10$

$h = -\frac{1}{40}(100)^2 + 10$

$h = -\frac{5}{2} + 10$

$\boxed{h = 7.5\text{m}}$

∴ The height is 7.5 m.

Homework (4.4) Pg. 186 #12-14

Answers: 1a) 80m b) 4s c) 8sd) 2s and 6s 2.a) 7m b)1s c)2m 3.a) 50 crates b)$15000 c) loses $210 4.a) 10m b) 20m c) 40m d) 7.5m

Anna Denisova Applications of Max/Min Nov. 15, 2022

STEPS

1. let statements
2. set up equations
3. perform substitution to get a single equation
4. Complete the square
5. Interpret the vertex
6. ∴ statement

e.g. You have 12m of fence. What dimensions will give you a maximum area if the shape fenced is a rectangle.

y is x? confusing
Try to NOT use x & y

let x represent the length (m)
let y represent the width (m)

① $2x+2y=12 \rightarrow x+y=6 \rightarrow x=6-y$

② $A=xy$

Sub x into ②

factored
$A=(6-y)y$
$A=6y-y^2$

standard
$A=-y^2+6y$
$A=-(y^2-6y)$

$\left(\frac{-6}{2}\right)^2$
$A=-(y^2-6y+9)+9$

vertex
$A=-(y-3)^2+9$
$V(\underset{y}{3},\underset{A}{9})$

Sub in $y=3$
$x=6-3$
$x=3$

$\boxed{\therefore \text{The dimensions are } 3m \times 3m}$

sum of squares a^2+b^2
square of sum $(a+b)^2$

e.g. The sum of two numbers is 16. Find the numbers if the sum of the squares is a min.

let a represent the first number
let b represent the second number

① $a+b=16 \rightarrow a=16-b$

② $M=a^2+b^2$

→

$(a+b)^2 = a^2 + 2ab + b^2$

① $a = 16 - b$

② $M = a^2 + b^2$

Sub a into ②

$M = (16-b)^2 + b^2$

$M = 256 - 32b + b^2 + b^2$

$M = 2b^2 - 32b + 256$

$M = 2(b^2 - 16b) + 256$

$M = 2(b^2 - 16b + 64) + 128$

$M = 2(b-8)^2 + 128$

$V(8, 128)$

$b \quad M$

sub b into ①

$a = 16 - 8$

$\boxed{a = 8}$

$\boxed{\therefore \text{The numbers are } 8 \text{ \& } 8}$

R Revenue - total income

P Profit - Revenue with costs deducted

p price - how much you sell each item for

q quantity - how many sold/made

C cost - how much it costs to produce each item

$\boxed{R = pq}$ $\quad P = pq - cq \quad P = R - C$

Eg. If a show charges \$1.60/person 200 people will buy a ticket. For every 40¢ increase in price, 10 less people will buy a ticket. What is the ticket price that will give the max revenu

Let x represent the number of increase in price

① $p = 1.6 + 0.4x$

② $q = 200 - 10x$

③ $R = pq$

Sub p and q into R

$R = (1.6 + 0.4x)(200 - 10x)$

$R = -4x^2 + 64x + 320$

$R = -4(x-8)^2 + 576$

$V(8, 576)$

$x \quad R$

Sub x into P

$P = 1.6 + 0.4(8)$

$P = 4.8$

$\therefore$ \$4.80 will give the max revenue.

Homework:

Worksheet #9, row 78